MOHAMMAD REHAN



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EXECUTIVE SUMMARY

Founder & Systems Architect at Rehan Quant Lab with 4+ years of hands-on experience building reliable commercial desktop software, document security tools, and high-performance offline systems. Creator of RQL DocumentShield™ (permanent, court-admissible PDF redaction software), Deterministic State Kernel (high-speed native Rust transaction engine with hardware SHA-256 logging and MCA Rule 11(g) compliance), and AegisDB (high-throughput in-memory time-series engine with sub-microsecond latency). Architect of 14 commercial desktop applications across 4 business categories, distributed globally on Gumroad with 78% enterprise users from the United States.

CORE COMPETENCIES & ARCHITECTURE STACK

- **Enterprise Document Privacy & Offline PDF Redaction:** Permanent removal of sensitive text, vector shapes, and font glyphs directly from PDF object streams (zero residual text, immune to layer-separation or copy-paste recovery), automated regex detection for Social Security Numbers (SSN), PAN cards, credit cards, and confidential PII, deep metadata stripping (EXIF, XMP), and court-admissible SHA-256 audit certification (HIPAA, GDPR Art. 17, ABA Opinion 06-442 compliant).
- **High-Performance Transaction Engine (Native Rust):** 100% offline, on-premises transaction ledger compiled into a native dynamic library (`dse_core.dll`). Delivers >240,000 writes/sec (4.7µs commit latency), sub-microsecond cached reads, hardware-accelerated SHA-256 Write-Ahead Logging (WAL), and rapid crash recovery replaying 25,000 transactions in under 77ms with zero data loss.
- **Commercial Desktop Software Architecture:** Built 14 standalone desktop applications across 4 business categories, packaged with custom Windows installers (`Setup.exe`), formal EULA terms, dedicated GUI uninstallers registered in Windows Settings (`HKCU`), clean high-DPI user interfaces, and complete 100% offline privacy.
- **Automated System Reliability & Risk Controls:** Automated pre-execution contract verification, stress-testing under simulated high-volatility market scenarios, sub-second exception recovery, strict capital preservation rules, and automated drawdown limits.
- **Low-Latency In-Memory Database (AegisDB):** High-throughput ring-buffer data structure (2.5M ticks/sec, 0.39µs latency, cacheline-aligned to prevent CPU false sharing), x86_64 CPU cycle measurement via RDTSC, and memory-mapped append-only Write-Ahead Logging with sub-5ms crash recovery.
- **SIMD Vectorized Financial Computing:** High-throughput financial boundary validation evaluating 10,000,000 operations (21.4M ops/sec, 46.7ns latency) using AVX2 SIMD vectorization, enforcing a 35% max drawdown ceiling, 30% reserve liquidity, and verified mathematical precision.
- **Multi-Interface Network Bonding & System Monitoring:** Developed RQL Conduit network link aggregation with real-time health heartbeats, automatic sub-10ms failover across redundant connections, and local sensor monitoring services.
- **Cloud Infrastructure & Systems Security:** AWS Mumbai (ap-south-1), Cloudflare Pages Edge deployment, Zero-Trust Network Access (ZTNA) via WireGuard/Tailscale private VPN mesh, and Linux systemd performance optimization.

KEY PRODUCTION SYSTEMS & ARCHITECTURES

RQL DocumentShield™ — Enterprise PDF Redaction & Metadata Sanitizer

- Architected an air-gapped PDF redaction engine that permanently removes underlying text, vector shapes, and font glyphs from raw PDF cross-reference tables (XREF), preventing copy-paste extraction, OCR discovery, or layer-separation recovery.
- Engineered automated high-speed multi-pattern regex scanners detecting US Social Security Numbers (SSN), Credit Cards (16-digit PAN), Employer Identification Numbers (EIN), and confidential contact coordinates across high-volume document bundles.
- Implemented deep metadata and XML XMP stream purges, stripping author credentials, modification histories, and hidden camera EXIF data.
- Built automated SHA-256 cryptographic verification generating court-admissible Certificates of Document Sanitization aligned with US HIPAA Safe Harbor (45 CFR § 164.514), EU GDPR Art. 17, and ABA Formal Opinion 06-442. Packaged with custom Windows installer and GUI uninstaller.

Deterministic State Kernel — High-Speed Native Rust Transaction Engine

- Engineered a commercial native Rust transaction engine compiled with Link-Time Optimization (`opt-level = 3`, `panic = "abort"`) producing a compact 1.06 MB C-ABI dynamic library (`dse_core.dll`) and 475 KB CLI tool (`dse-kernel-cli.exe`).
- Benchmarked write throughput exceeding 240,000 writes/sec (4.69µs commit latency) and read throughput exceeding 1,570,000 reads/sec (604ns cached latency).
- Implemented hardware SHA-256 Write-Ahead Logging with rapid crash recovery replaying 25,000 state records in 76.99ms with zero data loss, compliant with MCA Rule 11(g) audit trail requirements. Integrated Python `ctypes` FFI bridge achieving >124,000 writes/sec without GIL contention.

RQL Desktop Software Suite — 14 Commercial Applications Across 4 Business Categories

- Architected, compiled, and published 14 commercial offline desktop applications on Gumroad (`quantify17.gumroad.com`), achieving organic international distribution with 78% customer acquisition from the United States.
- **Document Privacy & Security:** Engineered *RQL DocumentShield*™ (permanent PDF redaction) and *Veritas Dossier*™ (statutory MCA Rule 11(g) audit trail logging).
- **Tax & Accounting Software:** Developed *TaxMatrix*™ *CA Reconciler* (automated GSTR-2B matching cross-referencing 100,000+ invoices in <1s) and *LedgerGuard*™ (automated Tally/SAP financial ledger reconciliation).
- **Infrastructure & Data Systems:** Built *Deterministic State Kernel* (Rust DLL >240k writes/sec), *AegisDB*™ (2.54M ticks/sec), *RQL Conduit*™ (multi-interface network failover), and *Industrial Sentinel*™ (sensor telemetry monitoring).
- **Retail POS & Billing Software:** Deployed *StoreFlow*™ *POS Workstation*, *ExpressBill*™ (thermal receipt POS), *StockAudit*™ (inventory barcode scanner), *RQL PriceTag*™ (shelf barcode sticker designer), and *LocalMesh*™ (offline multi-counter LAN synchronization).

AegisDB — High-Throughput In-Memory Database (Open-Source MIT)

- Architected an ultra-low-latency in-memory state engine and circular tick ring buffer with 64-byte cacheline padding achieving 2,535,916 ticks/sec at 0.39µs latency with zero false sharing across CPU cores.
- Implemented high-resolution CPU timing via x86_64 RDTSC instructions measuring execution performance in 38 cycles per operation.
- Evaluated 10,000,000 financial operations using AVX2 SIMD vectorization at 21.4M ops/sec (46.7ns per check) with verified mathematical precision.
- Built a memory-mapped (mmap) append-only Write-Ahead Log (WAL) achieving 3,559,925 syncs/sec (0.28µs append) and 3ms crash recovery with zero data loss.

Commercial Retail Point of Sale & Inventory Software — (StoreFlow™ & ExpressBill™)

- Engineered commercial offline Point of Sale (POS) and inventory ledger managing retail store operations, thermal barcode generation, and multi-counter LAN synchronization.
- Features multi-counter SQLite WAL concurrency protection, real-time margin analytics, and automated stock replenishment forecasting, reducing inventory holding costs by 30%.

PROFESSIONAL EXPERIENCE

Founder & Systems Architect • Rehan Quant Lab™ *January 2023 – Present | Patna, India (Remote Global)*

- Direct independent systems research and commercial software engineering across low-latency financial systems, deterministic transaction engines, and document privacy applications.
- Engineered and launched 14 commercial desktop applications on Gumroad (`quantify17.gumroad.com`), generating qualified organic traffic with 78% customer discovery from the United States.
- Architected `RQL DocumentShield™` ($199–$799) and `Deterministic State Kernel` ($249) as standalone high-margin commercial software releases.
- Directed end-to-end product architecture, native C/Rust compilation, PyMuPDF document engines, and international Cloudflare Pages deployment.

Store & Operations Manager • Ladies Zone Exclusive Showroom *March 2021 – Present | Patna, India*

- Direct day-to-day operations, inventory replenishment, and customer relationship management for a high-volume retail fashion showroom in Patna, Bihar.
- Architected and deployed in-house biometric attendance audit ledger, retail POS billing software, and custom business workflow automations.
- Bridged physical business requirements with automated software engineering to eliminate operational overhead and inventory discrepancies.
- Managed vendor relations, staff scheduling, commercial margin analysis, and customer dispute resolution with 100% operational continuity.

EDUCATION & ACADEMIC CREDENTIALS

Bachelor of Computer Applications (BCA) • National Business College *Graduated January 2025 | Patna, India*

- **Core Coursework:** Object-Oriented Software Design, Database Management Systems, Data Structures, Web Technologies, Network Security.
- **Capstone Project:** Enterprise Retail Management System with Real-Time Inventory Tracking and POS Integration.

TECHNICAL ENVIRONMENT & PRODUCTION SAFEGUARDS

- **Languages & Core Systems:** Rust (Native C-ABI FFI, LTO), C (AVX2 SIMD), Python 3.14/3.12, PyMuPDF, AsyncIO, FastAPI, Streamlit, Modern JavaScript.
- **Databases & Protocols:** AegisDB (In-Memory Ring Buffers), Deterministic State Kernel (Rust WAL), SQLite (WAL micro-journaling), RESTful APIs, WebSocket, WireGuard/Tailscale VPN.
- **Architectural Principles:** Fault-Tolerant Systems, Strict Capital Preservation, Permanent PDF Sanitization, 100% Offline Air-Gapped Privacy, Standalone Windows Executables.
- **DevOps & Infrastructure:** AWS Mumbai (ap-south-1), Cloudflare Pages Edge, Linux systemd (Nice=-10), Git, 250+ Unit & Integration Tests (100% Passed).